

Climate forcing due to tropospheric and stratospheric ozone

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Abstract. Radiative forcings, preindustrial to present, are presented due to changes in both tropospheric and total ozone. Unlike previous studies, we use a combination of present-day satellite data, ozonsondes, and a chemistry model to produce a hybrid monthly mean ozone data set from 1870 to 1990. We calculate the radiative forcing due to the prescribed ozone distribution using a three-dimensional climate model that employs seasonally evolving fixed dynamical heating in the stratosphere. We find that tropospheric ozone causes a seasonal fixed dynamical radiative forcing (SEFDH) of 0.30 W m^{-2} over the time period 1870–1990. The spatial pattern of forcing is correlated strongly with the change in tropospheric ozone. When we allow for observed changes in stratospheric ozone we calculate that the forcing due to total ozone from 1870 to 1990 is 0.29 W m^{-2} , with 0.18 from shortwave and 0.11 W m^{-2} from longwave processes. Thus reduction of stratospheric ozone since 1970 offsets the forcing by only -0.01 W m^{-2} . We also consider to what extent increased tropospheric ozone offsets the direct forcing by sulfate aerosols. We find that for June–July–August the sulfate forcing is -0.76 W m^{-2} , while the addition of ozone forcing reduces this to -0.40 W m^{-2} . Finally, we carry out sensitivity studies with regard to the uncertainty in preindustrial ozone levels.

1. Introduction

Radiative forcing of the climate system is an important measure of the effects of anthropogenic activity on the climate system [Intergovernmental Panel on Climate Change (IPCC), 1996]. This forcing is defined as the change in the net (longwave and shortwave) radiative flux at the tropopause. This change results from a human-influenced change in concentration of an atmospheric constituent from preindustrial conditions to some future time. The change in flux is calculated assuming all other climate variables are held fixed. Over the past 10 years, radiative forcings for a large number of greenhouse gases and certain aerosols have been evaluated for both the 20th and the 21st centuries [e.g., IPCC, 1996]. Of all of the greenhouse gases, perhaps the greatest uncertainty in forcing arises from ozone. Ozone forcing is particularly difficult to assess for a number of reasons.

First, the forcing is very sensitive to the vertical distribution [Lacis *et al.*, 1990] of ozone and to the exact specification of the tropopause height. Second, it exerts a forcing effect not only in the longwave spectral region (i.e., a greenhouse effect) but also absorbs in the shortwave spectral region. Third, it is common to calculate this forcing allowing for the thermal adjustment to stratospheric temperatures, usually assuming fixed dynamical heating (FDH) in this region [e.g., Forster *et al.*, 1997]. The exact manner in which this thermal calculation is carried out can differ among models. Finally, there are no comprehensive long-term global measurements of ozone. Unlike well-mixed

constituents such as carbon dioxide or methane, where we have in situ and ice core records of the temporal evolution of these species, ozone, because of its shorter lifetime, does not allow for an easy reconstruction of its changes in space and time. These sources of uncertainty have led to a large number of studies to better understand the role of ozone in forcing the climate system [e.g., Bernsten *et al.*, 1997; Brasseur *et al.*, 1998; Haywood *et al.*, 1998; Stevenson *et al.*, 1998]. To date, the forcing calculations have relied heavily on chemical transport model simulations of either tropospheric and/or stratospheric ozone.

The present study is unique in that we employ as much observational data as possible to define the distributions of tropospheric and stratospheric ozone. To define the ozone distribution for preindustrial times, we must rely on an atmospheric chemistry model. This hybrid approach for determining the ozone forcing constrains the trend and distribution in atmospheric ozone to observations and thus should yield more accurate forcings. The study is organized in the following manner: Section 2 discusses the sources for the ozone distributions, section 3 describes the radiative model used for the forcing calculations, section 4 presents the results for the forcing for various time periods, and section 5 summarizes our results and discusses further areas for improvement in calculating ozone forcing.

2. Ozone Data Sources

The focus of this study is to estimate the radiative forcing between the preindustrial period and the present. Thus ozone distributions for the times 1870 and 1990 will be emphasized here. However, monthly ozone data sets for the entire time period between 1870 and 1990 have been created for use in climate model simulations of the 20th century. This section describes the methods used to create these ozone data sets for this entire time period.

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The overall strategy is to use present-day maps of tropospheric ozone column and total ozone trends inferred from satellite measurements along with a model-based estimate for the preindustrial (1870) ozone. From these estimates, seasonal maps of the tropospheric ozone column changes are obtained. The profile for these changes along with their time rate of change complete the specification.

Present-day (1990) tropospheric ozone column is estimated from the tropospheric residual maps [Fishman *et al.*, 1997], which are produced by subtracting the stratospheric column measured by SAGE from the total column measured by TOMS (using the Version 7 retrieval) using about 10 years of data. Limited spatial sampling of the SAGE occultation data requires that a large amount of data be averaged to produce mean maps of the tropospheric column. The residual total ozone maps are seasonal and available from 50° north to south. Poleward of 50° we apply the change in ozone at 50° to the model ozone field at all latitudes above 50°. Thus only the change in ozone is extrapolated beyond 50°. Clearly this is a limitation to obtaining global coverage.

The residual ozone maps are shown [Fishman *et al.*, 1997] to be accurate to about 20% when compared to ozonesonde data, with the residual technique tending to underestimate the ozone. Preindustrial (1870) tropospheric ozone is computed here with the Garcia-Solomon two-dimensional model [Garcia and Solomon, 1994]. The model contains detailed representations of ozone photochemistry and stratospheric dynamics and produces a realistic total ozone field. Loss at the surface is modeled by a mixing ratio boundary condition at the surface of 10 and 25 ppbv in the Northern and Southern Hemispheres, respectively. The Northern Hemisphere value is taken from ground-based measurements [Marenco *et al.*, 1994], while the larger value adopted in the Southern Hemisphere reflects the smaller surface sink there due to decreased landmass. For comparison purposes the model was also run with surface boundary conditions more representative of clean locations in the 1970s. Comparisons between the model and the seasonally averaged ozonesonde data at clean Canadian (Edmonton, Goose Bay, and Churchill) and Australian (Aspendale) stations in the 1970s (which are relatively free of local pollution sources) show that the model produces an accurate vertical gradient of ozone in the troposphere and lower stratosphere and supposedly gives the preindustrial model profiles. A preindustrial tropical biomass burning component was added to the zonal mean model results to account for natural burning. This was estimated by taking 25% of the tropical tropospheric residual ozone after removal of the zonal minimum [see Portmann *et al.*, 1997]. The estimated tropospheric column ozone perturbation from 1870 to 1990 is obtained by subtraction of the 1870 model estimate from the present-day residual ozone maps.

In addition to the column ozone perturbation maps, the vertical profile of the perturbations is needed. In order to obtain insight on this, the differences between ozonesonde data obtained at polluted and clean locations were examined. For example, the seasonal averages obtained at the relatively clean Canadian and Australian stations were subtracted from similar averages at Hohenpeissenberg and Payerne, which are in polluted regions. These differences suggest that to a reasonable approximation the ozone perturbation is constant with altitude (in mixing ratio) in the free troposphere. Thus a constant mixing ratio perturbation profile is adopted up to the tropopause, which is assumed to vary with latitude using a simple formula. Although more localized perturbations are

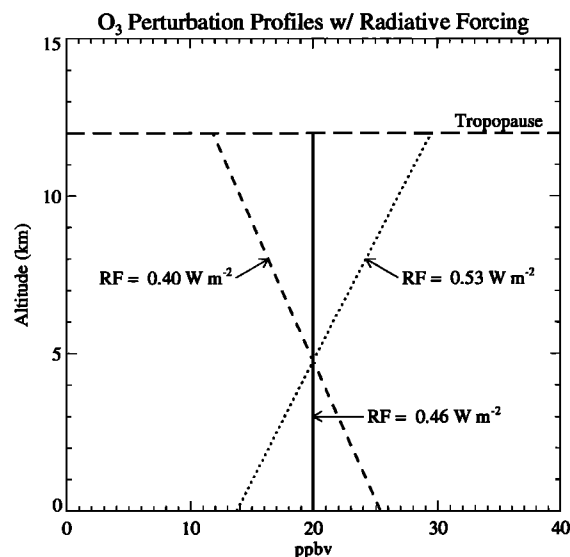


Figure 1. Vertical profiles of three linear tropospheric ozone perturbations (ppbv). Change in the longwave tropopause flux (W m^{-2}) due to each of these perturbations is noted in the figure. All three perturbations lead to a 12.7 DU change in total tropospheric ozone.

expected close to polluted regions, these are mixed throughout the troposphere over broader spatial scales.

To estimate the sensitivity to this choice, the radiative forcing of a constant perturbation of an increase of 12.7 Dobson units (DU) was compared to those with a constant slope where the tropopause value was one half and twice the surface value but which preserves the tropospheric column (Figure 1). These cases show only a $\pm 15\%$ change in radiative forcing. This relatively small change from dependence on the profile is due to the perturbation being constant in mixing ratio, which emphasizes changes over a broad region of the middle troposphere, compared with those constant in density, which emphasize changes close to the tropopause [Lacis *et al.*, 1990; Forster and Shine, 1997].

At northern middle latitudes it is reasonable to assume that the growth in tropospheric ozone increased with industrial growth and that this roughly scaled with carbon dioxide emissions. We recognize that this is not absolutely correct, since tropospheric ozone is tied to ozone precursors like NO_x , which has different emissions compared to carbon dioxide. The carbon emission scenario is similar to the IPCC [1995] data and was used to produce a normalized function that ranges from zero to 1 from 1870 to 1990. It is important to recognize that this function plays no role in the results of this study, since we focus on the two end points of this function, i.e., the preindustrial and the present.

Stratospheric ozone changes are assumed to begin in 1970, since ground-based measurements of total ozone show that a significant ozone decline began during the 1970s [World Meteorological Organization (WMO), 1995]. The implemented stratospheric ozone changes are based on satellite estimates of the decadal trends in total ozone, which are dominated by changes in the stratosphere. Zonal mean trends inferred from the Nimbus 7/Meteor 3 TOMS data set over the period 1979–1994 are used [McPeters *et al.*, 1996], which vary with month and latitude and provide nearly global coverage. No data are available during polar night, and so the trends there are as-

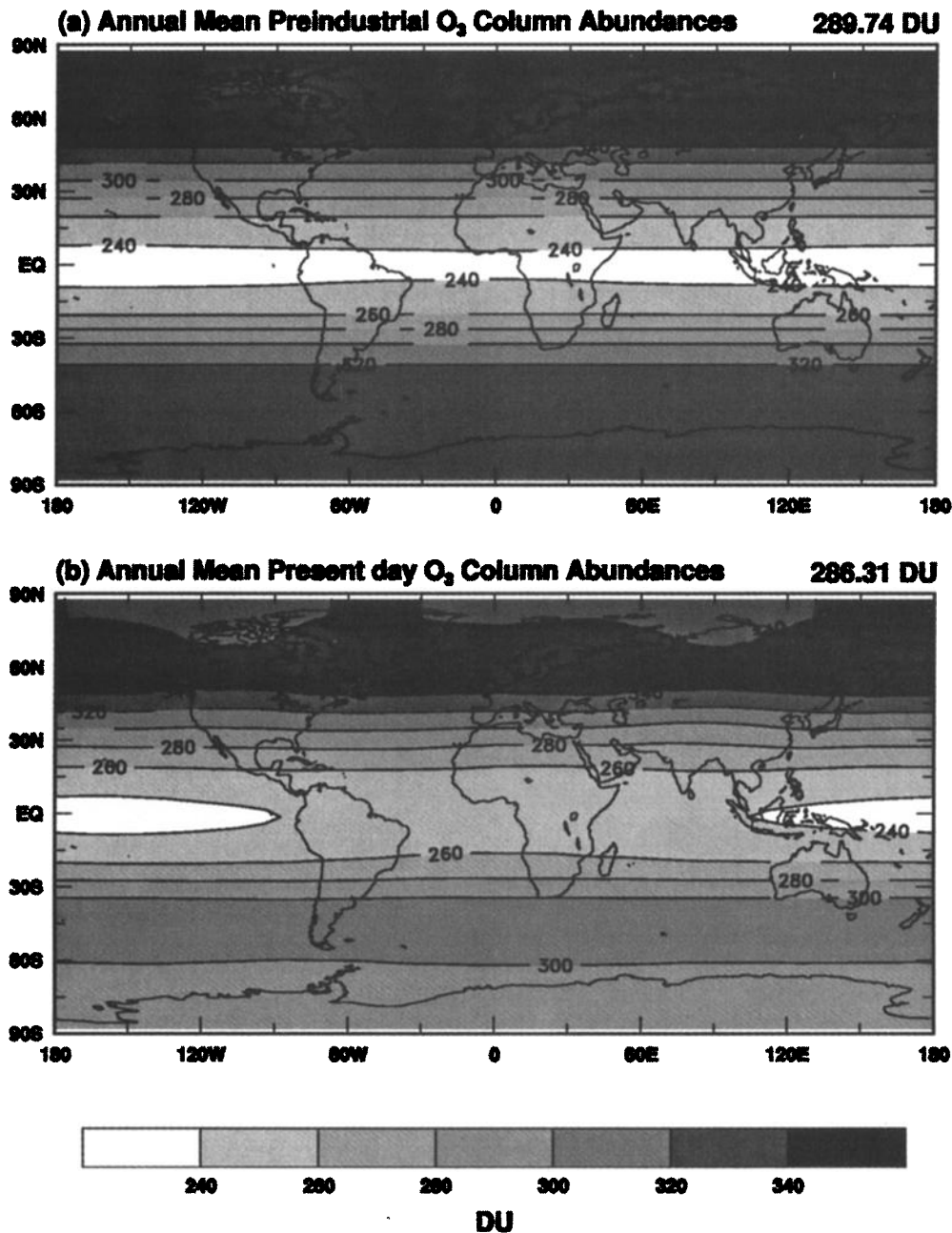


Figure 2. Annual mean geographic distribution of total column abundance of ozone in Dobson units (DU) for (a) 1870 and (b) 1990 time periods.

sumed to be equal to the value entering polar night at that latitude. No adjustment of the trends for changes in the troposphere was made, since these are believed to be relatively small in the zonal mean over this time period over most of the globe [WMO, 1995]. The inferred trends are used directly during the 1980s and are reduced by 50% during the 1970s. This scaling is based on the relative magnitude of ground-based estimates of the trends during the 1970s and 1980s [WMO, 1995].

The ozone profile changes play an important role in the magnitude of its radiative forcing. In general, changes closer to the tropopause play an increased role since changes in local temperature affect infrared emission into the troposphere. Thus employing reasonable profiles of the perturbation is important. We have used estimates of the slope of the profile of ozone trends inferred from ozonesonde data below 30 km and

Umkehr data from 30 to 50 km [Miller *et al.*, 1995]. These are seasonal below 30 km and annual above. The trend profiles are scaled to produce the total ozone changes for a given latitude, season, and year. The stratospheric trends are set to zero below the latitudinally varying tropopause. The peak ozone trend in the upper stratosphere near 40 km is limited to -10% per decade, based on estimates inferred from SAGE data [Wang *et al.*, 1996]. Thus in locations with large column ozone changes, the largest changes are implemented in the lower stratosphere. Since the midlatitude lower stratospheric profile trends in winter and spring are very similar in shape (not magnitude) to the profile changes at high latitudes (including the ozone hole), separate profile slopes were not used in these locations. In the tropics the trends are assumed to be zero below 30 km, and the Umkehr profile slope is used above.

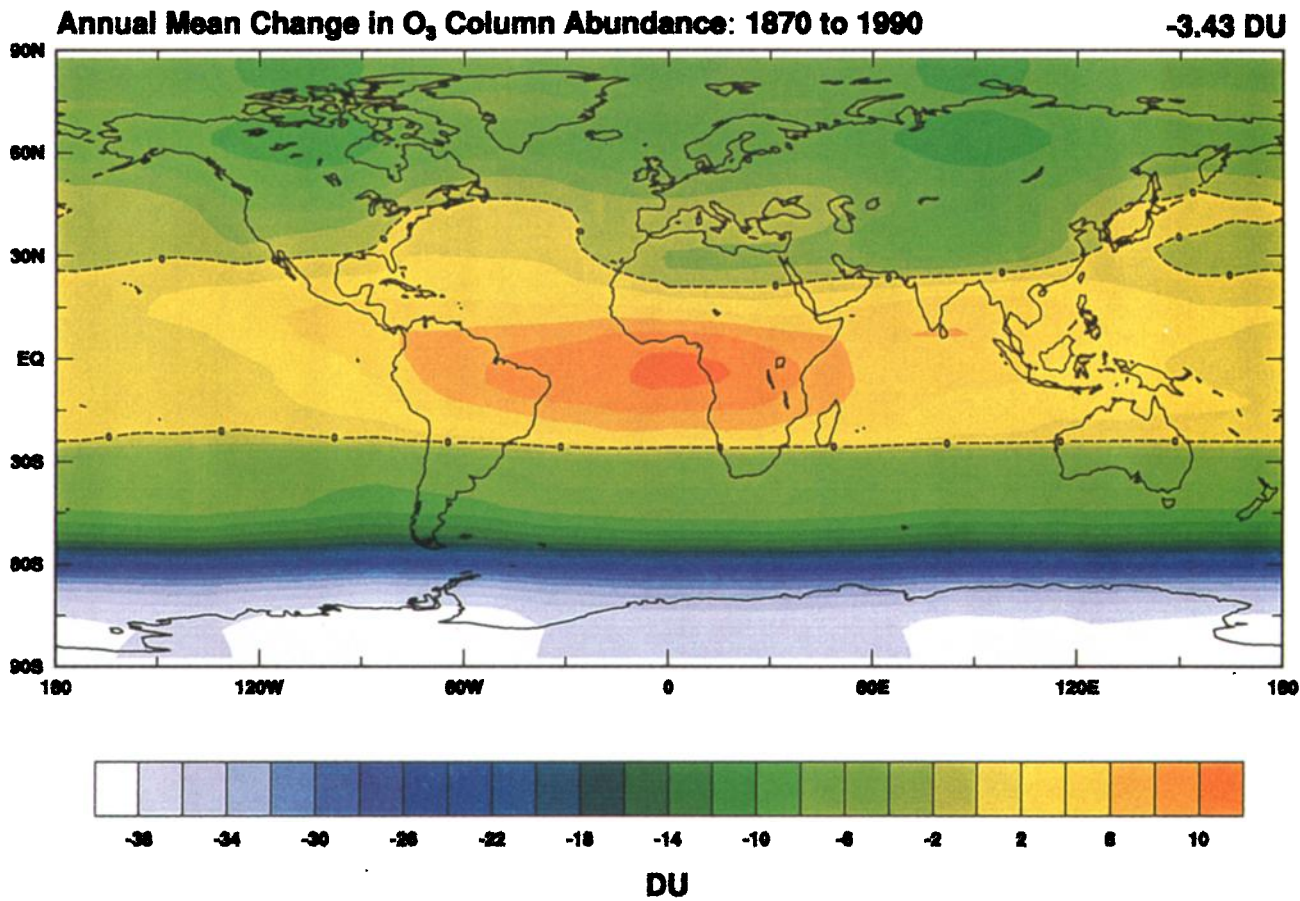


Plate 1. Annual mean geographic distribution in the change in column ozone (DU) from 1870 to 1990.

Total annual mean column ozone amount in Dobson units (DU) for 1870 and 1990 are shown in Figure 2. The 1870 distribution is mainly zonal in structure, since it is based on the two-dimensional chemistry model. Minimum ozone occurs in the tropics with levels below 235 DU. Ozone levels in the Northern

Hemisphere are higher by ~20 DU due to the asymmetry in dynamical circulation between the hemispheres. The total column ozone for 1990 conditions, Figure 2b, contains more zonal asymmetry, since this field is constrained by the TOMS observations. Again, the hemispheric asymmetry is due mostly to

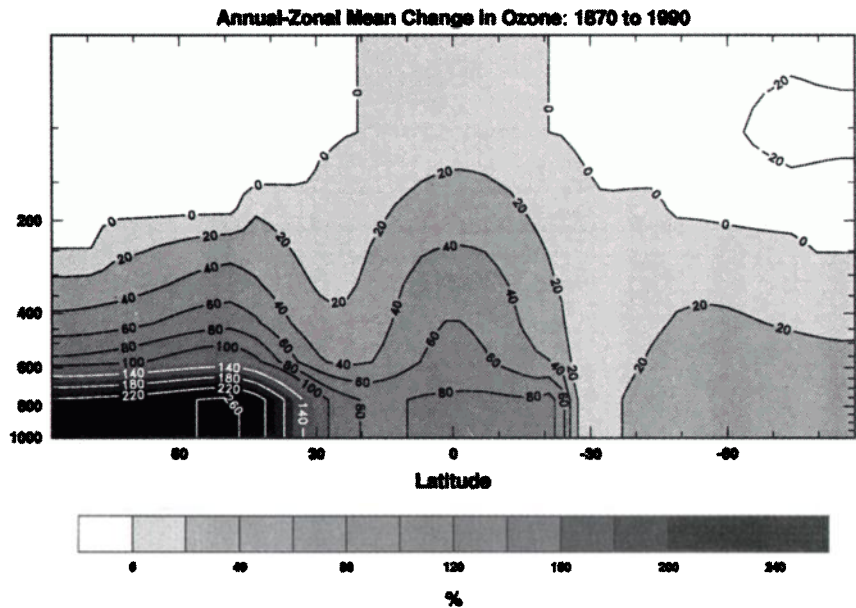


Figure 3. Annual mean, zonal mean of percent change in ozone from 1870 to 1990.

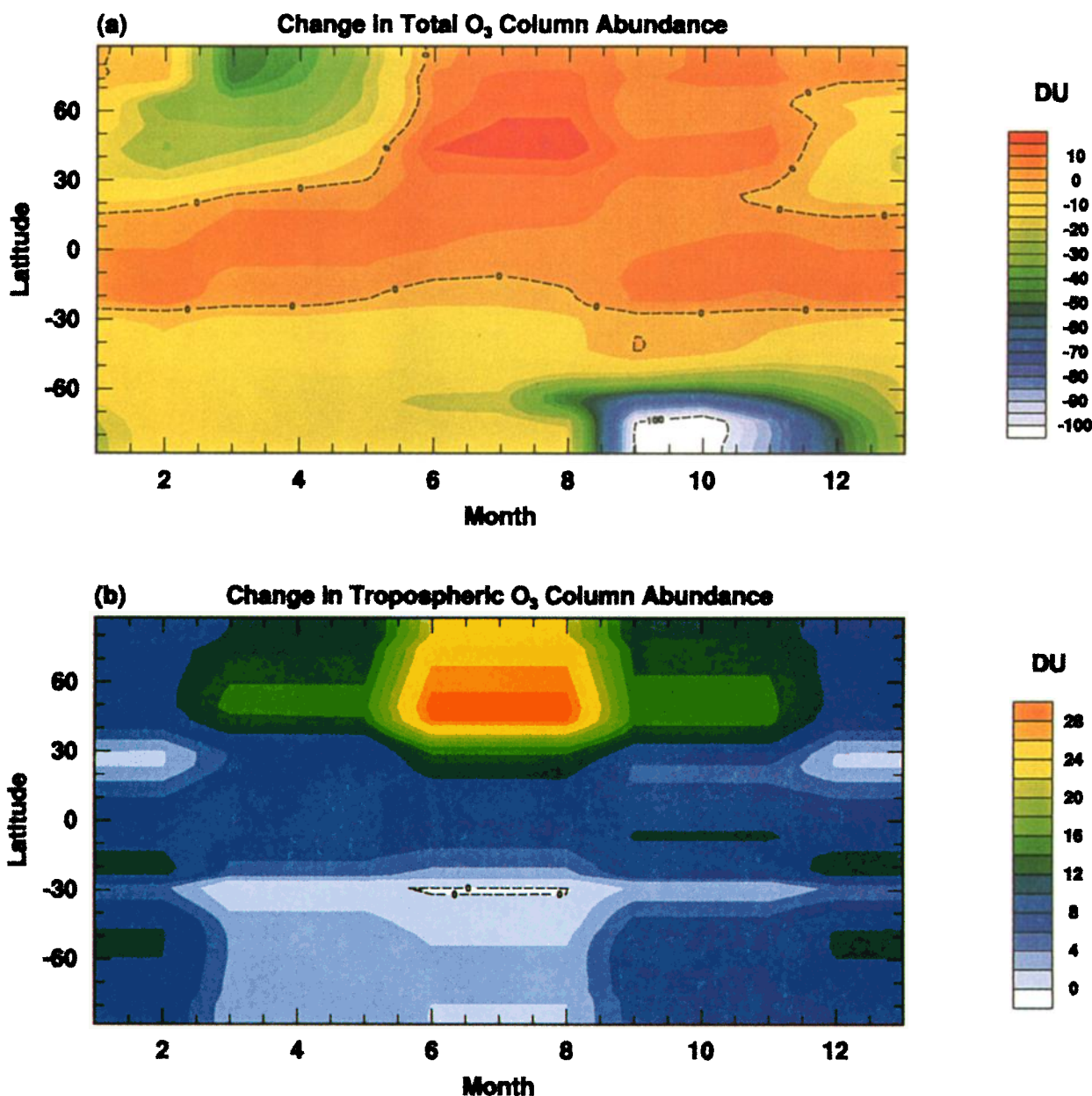


Plate 2. Seasonal cycle of the zonal mean change in (a) total ozone and (b) tropospheric ozone in DU from 1870 to 1990.

hemispheric differences in dynamical circulation. However, superimposed on this natural source of asymmetry is the Northern Hemisphere perturbation due to anthropogenic activity.

The annual mean geographic distribution for the change in column ozone (DU) from 1870 to 1990 is shown in Plate 1. Depletion of stratospheric ozone dominates the column abundance change in the Northern and Southern Hemisphere extratropics. Note that the largest depletions in ozone in the Northern Hemisphere occur over the North American continent and the central Asian region. Change in column ozone in the Northern Hemisphere is as low as -8 DU, while the decrease in the Southern Hemisphere is dominated by the Antarctic ozone hole. In the tropics the column is dominated by increases in tropospheric ozone associated with biomass burning [Fishman and Larsen, 1987; Portmann et al., 1997; Hudson and Thompson, 1998]. The global annual mean change in total column ozone is -3.43 DU. The annual zonal mean

percent change in ozone from 1870 to 1990 is shown in Figure 3. Below 600 mbar, between 30° and 60° north, ozone increases between 140 and 260%. In the tropical lower troposphere, ozone increases of 80–100% are inferred. Depletion of stratospheric ozone occurs in both hemispheres at levels of zero to 20%. Depletion associated with the Antarctic ozone hole is larger than 20% in the annual mean and is nearly complete in austral spring (October) at some altitudes.

The seasonal cycle of the zonal mean change in total column ozone, 1870–1990, is shown in Plate 2a. From late November to May, poleward of 15° north, the total column ozone is negative, with reductions as large as -30 to -50 DU at high latitudes. From June to October the ozone change is positive, with increases reaching 20 DU at midlatitudes. This increase is associated with increased photochemical production of tropospheric ozone (see Plate 2b). In the tropical belt, between 15° north and 25° south, the ozone change is positive, with a max-

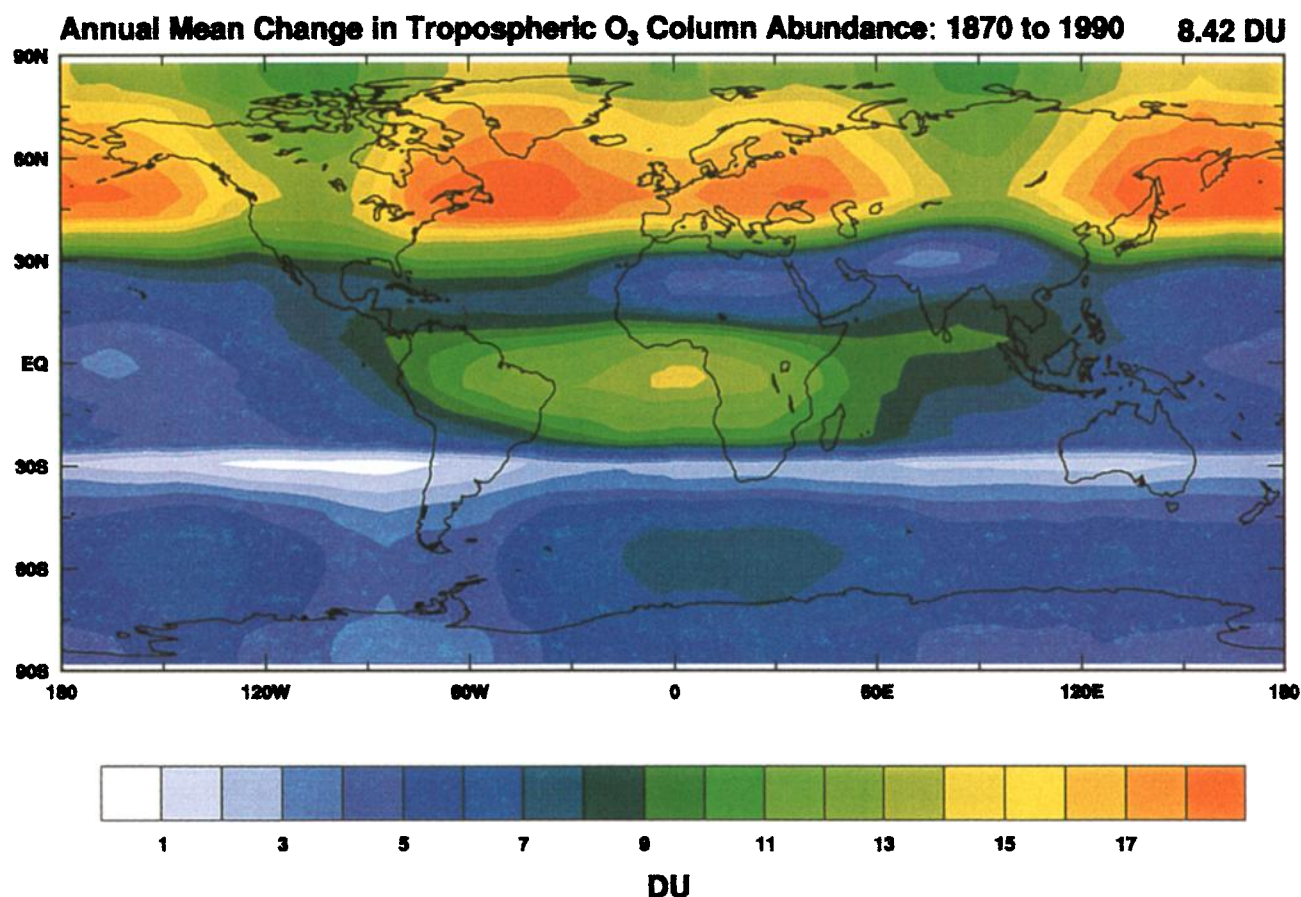


Plate 3. Annual mean geographic distribution of change in tropospheric ozone (DU) from 1870 to 1990.

Table 1. Vertical Level Structure of the Seasonal Fixed Dynamical Heating Model

Level Number	Pressure, mbar
1	0.191952
2	0.2554328
3	0.3399075
4	0.452319
5	0.6019067
6	0.8009644
7	1.065853
8	1.418343
9	1.887407
10	2.511596
11	3.34221
12	4.447522
13	5.918371
14	7.875647
15	10.48022
16	13.94616
17	18.5532
18	24.69579
19	32.86298
20	43.73117
21	58.1936
22	77.43894
23	103.0489
24	137.1285
25	182.4785
26	242.8264
27	323.132
28	429.9958
29	572.2006
30	761.4344
31	1000.000

Levels are listed in millibars.

imum in September through December, associated with seasonal biomass burning. In the Southern Hemisphere, poleward of 25° south, the change in ozone is negative for all months. The Antarctic ozone hole is evident in austral spring with reductions greater than 100 DU.

Since many studies have calculated the radiative forcing due to changes in tropospheric ozone [e.g., Brasseur *et al.*, 1998; Stevenson *et al.*, 1998], we consider changes in tropospheric ozone alone. The geographic distribution of the change in annual mean tropospheric ozone between 1870 and 1990 is shown in Plate 3. The largest increases occur in the Northern Hemisphere associated with the industrial production of ozone precursors (e.g., NO_x, CO). Column abundance increases exceeding 17 DU occurs off the east coasts of China and the United States. The second largest increase in tropospheric ozone, ~15 DU, occurs in the tropical Atlantic associated with

Table 2. Trace Gas Mixing Ratios and Sulfate Burden for Preindustrial and Present Conditions

Constituent	PreIndustrial	Present
CO ₂	280 ppmv	353 ppmv
CH ₄	0.800 ppmv	1.72 ppmv
N ₂ O	0.288 ppmv	0.31 ppmv
CFC-11*	0	0.280 ppbv
CFC-12	0	0.484 ppbv
Sulfate Burden	0.74 mg m ⁻²	2.97 mg m ⁻²

*Mixing ratio of CFC-11 is multiplied by a factor of 2.01 to account for the presence of CFCs and halons.

biomass burning. The global annual mean increase in ozone from 1870 to 1990 in tropospheric ozone is 8.42 DU. The seasonal cycle in zonal mean change in tropospheric ozone from 1870 to 1990 is shown in Plate 2b. The field is dominated by increased tropospheric ozone at Northern Hemisphere midlatitudes mainly from April to November. The slight decrease at 30° south, June through August, is due to a small uncertainty in determining the height of the model tropopause.

3. Model Description

Radiative forcing is calculated from a modified version of the NCAR (National Center for Atmospheric Research) CCM3 radiation model [Kiehl *et al.*, 1998a]. The model is global as with the CCM3, but performs only radiation calculations. The temperature and moisture profiles of the model atmosphere are prescribed based on data from the ECMWF analyses [Kiehl and Briegleb, 1993]. Cloud properties were obtained from a simulation with the NCAR CCM. These include cloud fraction, liquid water path, and water phase. The CCM cloud fraction at each model level was renormalized by a single global value to insure a global mean cloud cover that agrees with International Satellite Cloud Climatology Project (ISCCP) observations [see Kiehl and Briegleb, 1993]. Surface albedo is also spatially varying and is determined from the

CCM surface boundary data. The radiation model is the same as used in the CCM3 [see Kiehl *et al.*, 1996, b]. In the longwave part of the spectrum the model employs a broadband emissivity/absorptivity scheme. The gases included are H₂O, CO₂, O₃, CH₄, N₂O, CFC-11, and CFC-12. CFC-11 is scaled to account for the following gases: CFC-113, CFC-114, CFC-115, CCl₄, CH₃CCl₃, HCFC-22, HCFC-141, HCFC-142, HCFC-123, halon 1211, halon 1301, halon 2402, CH₃Br, CH₃Cl, and HFC-134a. The model does not include the radiative effects of the weak 14 μm band of ozone. Shine *et al.* [1995] have shown that this band contributes an additional -0.025 W m^{-2} forcing due to stratospheric ozone depletion. The shortwave model is based on a δ -Eddington scheme that uses 18 spectral intervals [see Briegleb, 1992]. The model accounts for absorption due to O₃, CO₂, H₂O, and O₂. Scattering due to aerosols and cloud droplets are also included. The aerosol used in the present calculations is the same as that described by Kiehl *et al.* [1999] and is determined from a sulfur cycle model in CCM3.

The model horizontal resolution is $\sim 2.8^\circ$ by 2.8° (T42 spectral resolution); the model employs 31 levels in the vertical, unlike the CCM3 which has only 18 levels. The vertical levels are listed in Table 1. The additional levels are required to accurately capture the radiative flux profiles near the upper troposphere and lower stratosphere. Above the tropopause the model solves for the stratospheric thermal structure using the assumption of seasonally evolving fixed dynamical heating

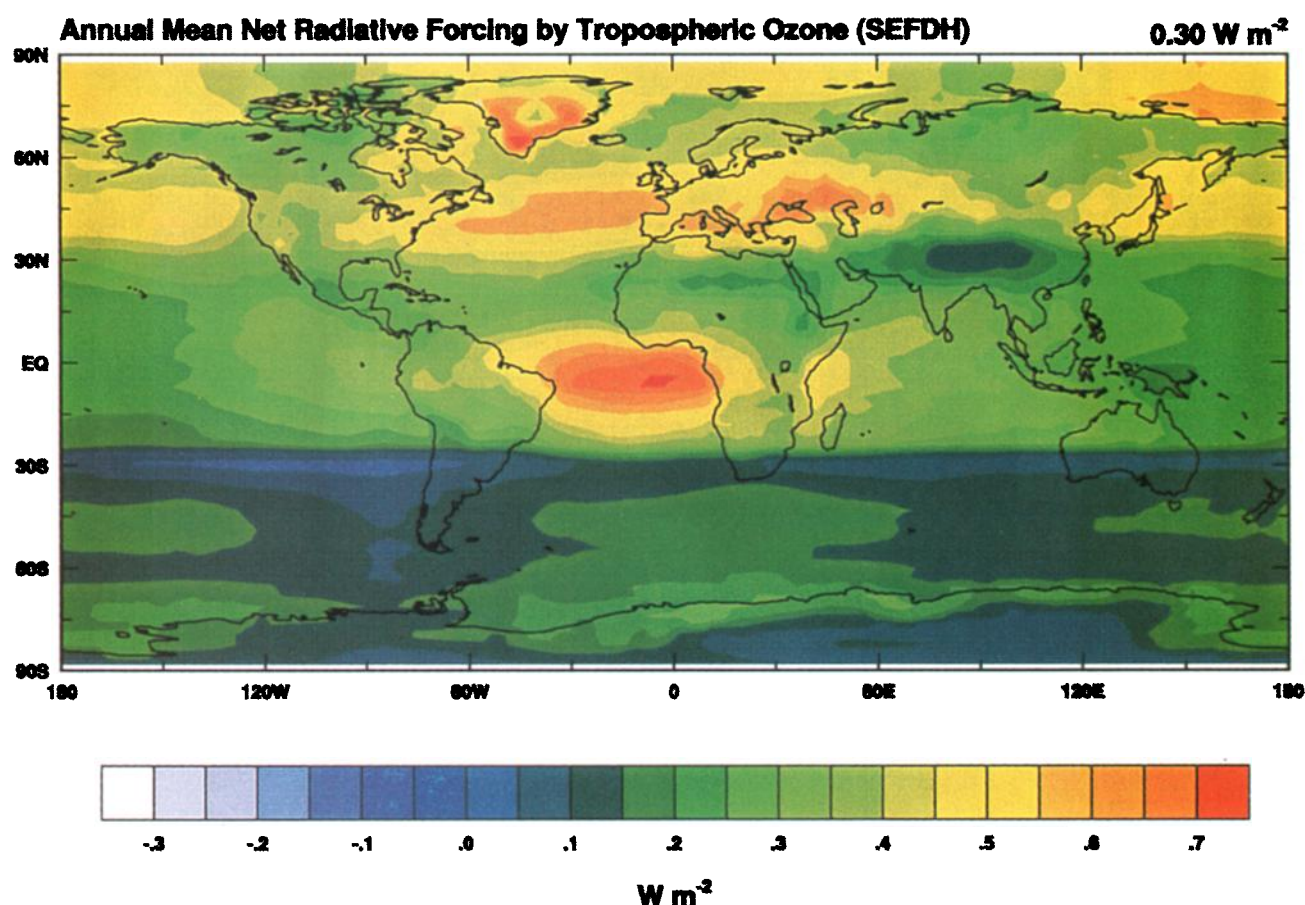


Plate 4. Annual mean geographic distribution in net (longwave plus shortwave) SEFDH forcing (W m^{-2}) due to changes in tropospheric ozone from 1870 to 1990.

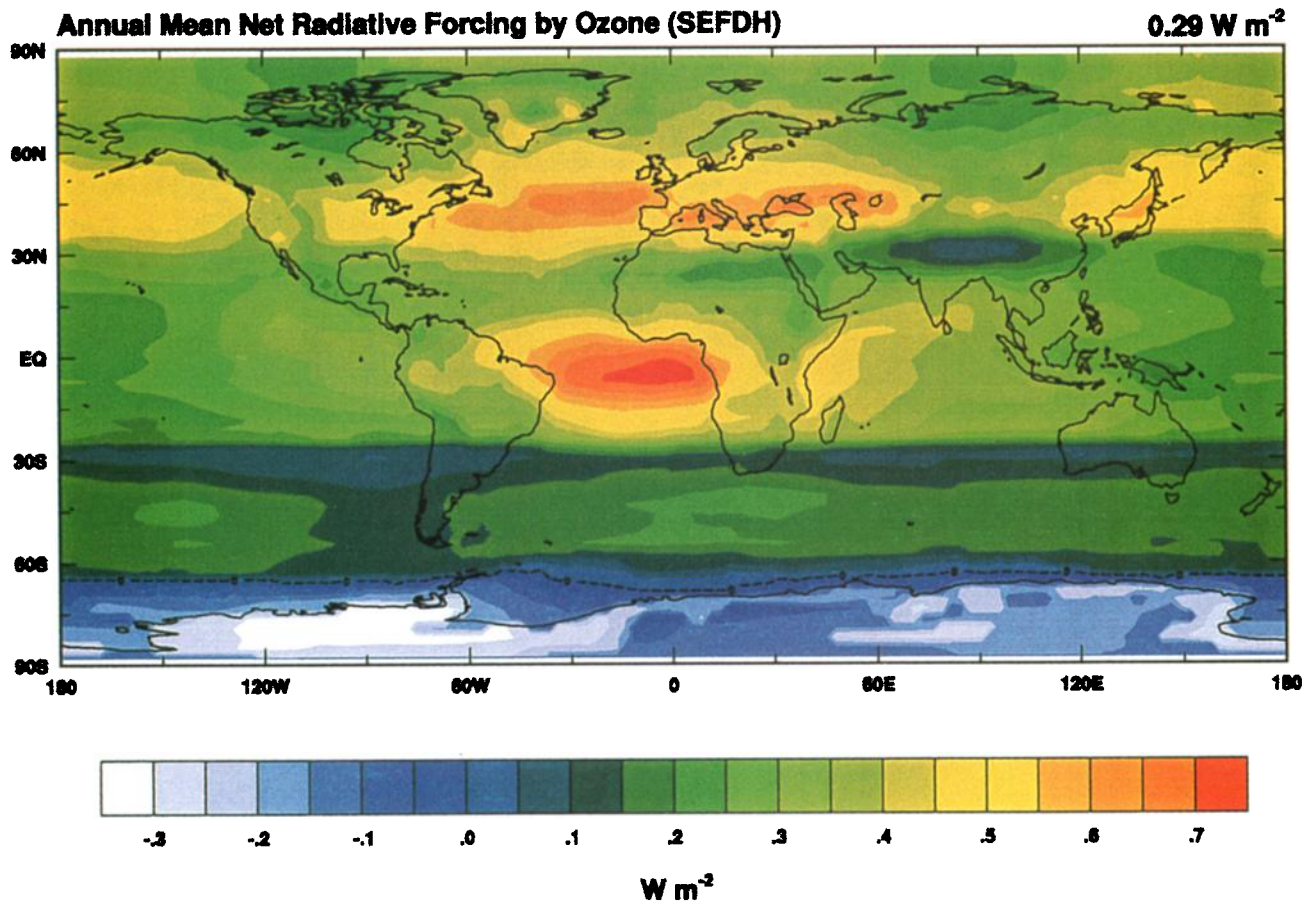


Plate 5. Annual mean geographic distribution in net (longwave plus shortwave) SEFDH forcing (W m^{-2}) due to changes in total ozone from 1870 to 1990.

(SEFDH) [see Forster *et al.*, 1997]. This approach uses observed temperatures and the column radiation model to calculate the dynamical heating at each grid point on the basis of the thermodynamic equation

$$Q_D = \frac{dT_0}{dt} - Q_R. \tag{1}$$

where Q_R is the net radiative heating rate, Q_D is the dynamical heating rate, and dT_0/dt is the observed temperature tendency (all quantities in K d^{-1}). Dynamical heating rates are calculated for present-day temperature conditions. This formulation implies that for present-day ozone the model will reproduce the observed thermal structure of the stratosphere. When the

Table 3. CCM3 Column Sensitivity Calculations Using Ozone Profile Data From Shine *et al.* [1995]

	Tropospheric O ₃			Stratospheric O ₃			Total O ₃		
	SW	LW	Net	SW	LW	Net	SW	LW	Net
<i>Instantaneous Radiative Forcing, (W m⁻²)</i>									
Mean	0.022	0.154	0.176	0.264	−0.066	0.199	0.287	0.089	0.376
s.d.	0.005	0.008	0.012	0.030	0.013	0.034	0.031	0.012	0.034
Min	0.017	0.142	0.163	0.229	−0.091	0.158	0.246	0.072	0.323
Max	0.031	0.168	0.192	0.308	−0.050	0.258	0.328	0.107	0.428
CCM3	0.025	0.136	0.161	0.239	−0.070	0.169	0.264	0.067	0.331
<i>Fixed Dynamical Heating Forcing, (W m⁻²)</i>									
Mean	0.022	0.112	0.134	0.265	−0.409	−0.144	0.287	−0.294	−0.008
s.d.	0.005	0.014	0.016	0.033	0.020	0.035	0.034	0.027	0.040
Min	0.017	0.100	0.119	0.228	−0.437	−0.190	0.245	−0.335	−0.066
Max	0.031	0.134	0.158	0.309	−0.379	−0.087	0.329	−0.248	0.043
CCM3	0.025	0.106	0.131	0.239	−0.426	−0.187	0.264	−0.321	−0.057

Forcing (W m^{-2}) is calculated for instantaneous and fixed dynamical heating. Mean is the mean of the nine models that participated in the intercomparison; s.d. is the standard deviation about the mean; Min is the minimum forcing of the models; Max is the maximum forcing of all models; CCM3 is the forcing obtained from the radiation model used for all of the present calculations.

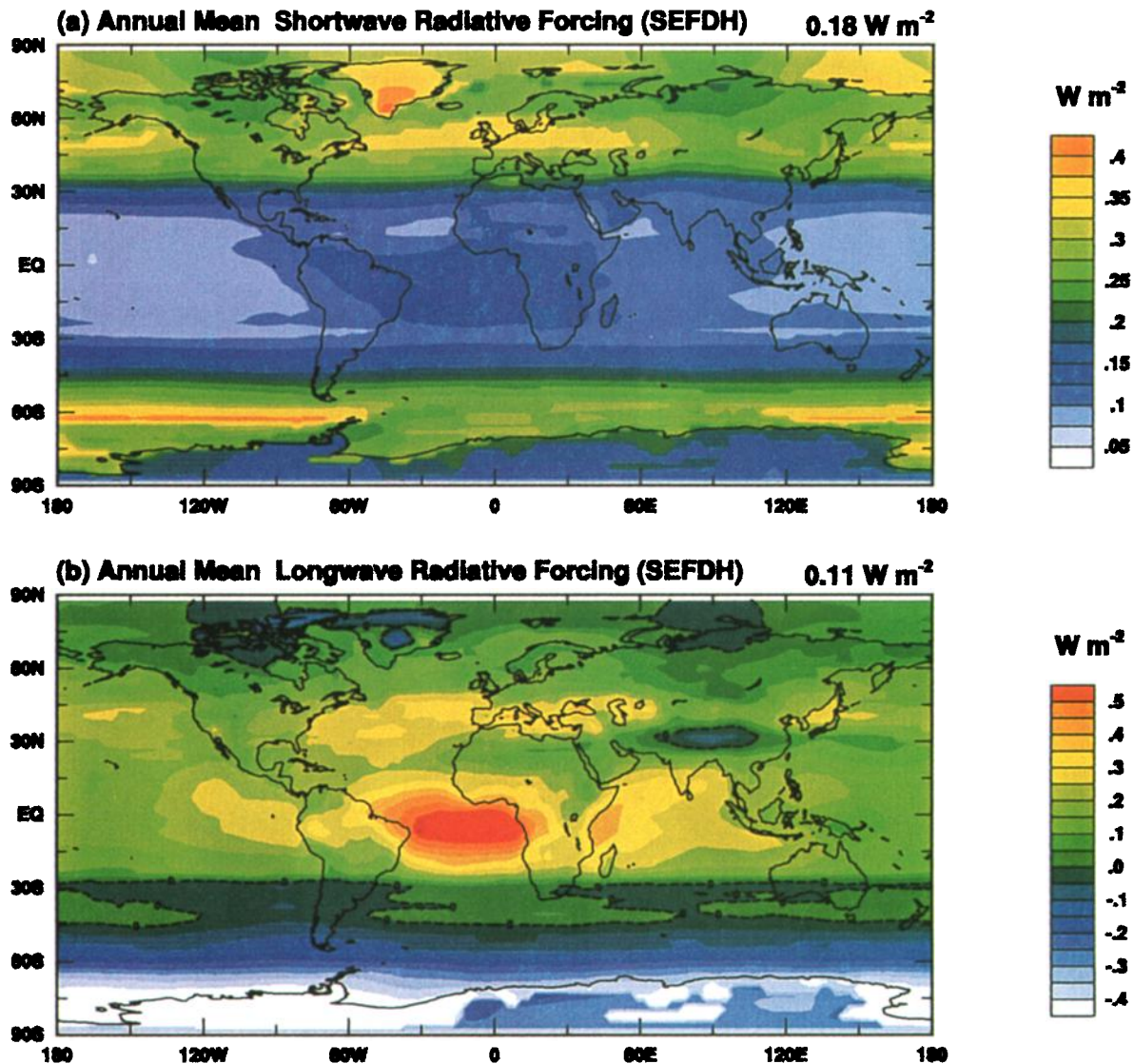


Plate 6. Annual mean geographic distribution in shortwave SEFDH forcing (W m^{-2}) and (b) longwave SEFDH forcing (W m^{-2}) due to changes in total ozone from 1870 to 1990.

ozone is perturbed, the thermodynamic equation is solved for a new temperature structure. The model is integrated through the annual cycle with a time step of 1 day, using the daily values of Q_D for the dynamical heating term.

The model is integrated for preindustrial conditions and for present-day conditions. The preindustrial and present trace gas concentrations are given in Table 2. The trace gas concentrations are based on the IPCC IS92a scenario. The CFC and halogen mixing ratios and radiative forcing per molecule are from the WMO [1998] and IPCC [1995] assessments. Radiative forcing is calculated at the tropopause. The tropopause level is determined by searching for a lapse rate of -3 K/km at each model grid point above 4 km (to avoid low-level tropospheric inversions) (B. A. Boville, personal communication, 1999). This definition is somewhat different from the WMO definition. We have carried out sensitivity studies with the WMO definition and find this has little effect on forcing. The forcing at this level is defined as

$$\text{Forcing} = -\Delta F_{\text{Trop}}^{\text{Net}} = \Delta F_{\text{Trop}}^{\downarrow} - \Delta F_{\text{Trop}}^{\uparrow} \quad (2)$$

where $F_{\text{Trop}}^{\downarrow}$ is the downward flux (shortwave and longwave) from the stratosphere into the troposphere, and $F_{\text{Trop}}^{\uparrow}$ is the

upward flux from the troposphere into the stratosphere. Delta (Δ) denotes a change due to a change in ozone, present minus preindustrial. Depletion of stratospheric ozone decreases the longwave flux to the tropopause but increases the shortwave flux into this region. Increased tropospheric ozone decreases the upward longwave flux into the stratosphere and increases the absorbed shortwave flux in the troposphere.

We have used ozone profiles from the Shine *et al.* [1995] intercomparison to compare the present model to nine other models used in that study. The profiles include a tropospheric perturbation, a stratospheric perturbation, and a total ozone perturbation. Calculations were carried out for instantaneous and fixed dynamical heating assumptions. The results from the present model and a summary of the nine other models are shown in Table 3. The CCM3 model is in good agreement with the other models. Considering the FDH results for the troposphere, the CCM3 shortwave and longwave results fall within the range of the other models. For the stratosphere-only results the CCM3 shortwave and longwave results are again within the range of the other models. For the total ozone case, the same conclusions hold.

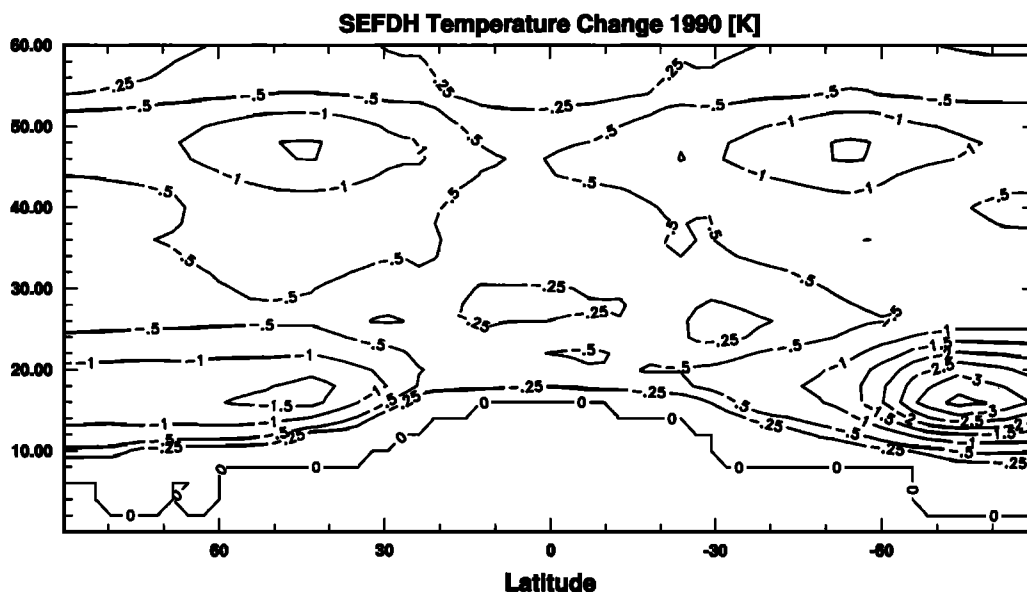


Figure 4. Annual mean, zonal mean change in stratospheric temperatures (K) from 1870 to 1990. Results are from the SEFDH model. Contour interval is 0.5 K for temperature changes less than -0.5 K, 0.25 K for temperature changes greater than -0.5 K.

4. Radiative Forcing

4.1. Tropospheric Ozone

A number of studies have focused on changes in tropospheric ozone by itself, thus we first consider the radiative forcing that has occurred due to changes in tropospheric ozone from 1870 to 1990. The geographic distribution of annual mean SEFDH net (longwave plus shortwave) radiative forcing for this period is shown in Plate 4. The geographic distribution of forcing correlates well with that of change in tropospheric ozone (Plate 3). Maximum forcing occurs in the Northern Hemisphere off the coasts of North America and Asia. Forcing in central Europe reaches 0.70 W m^{-2} . The forcing over the southern tip of Greenland is due to enhanced shortwave absorption over a high-albedo surface. In the tropics the largest forcing associated with biomass burning occurs over the central equatorial Atlantic Ocean. Forcing in the Southern Hemisphere is less than 0.15 W m^{-2} . Note there is no negative forcing in Plate 4, we use the same scale in this figure as in Plate 5. The strong spatial correlation between forcing and ozone change is not produced in other studies [e.g., Haywood *et al.*, 1998; Bernsten *et al.*, 1997]. These studies predict maximum ozone forcing over the Sahara Desert. They attribute this to the existence of ozone over this region, in conjunction with a large tropopause to surface temperature gradient, which results in a large greenhouse effect. The ozone distributions used in these studies came from chemical transport simulations. Comparison of our observed change in tropospheric ozone (i.e., Plate 3) with these model simulations (for example, Figure 2 of Haywood *et al.* [1998]) indicates that the models predict too much tropospheric ozone over the Saharan region ($10\text{--}15 \text{ DU}$ compared to less than 5 from the satellite data). Thus we argue that the actual spatial distribution of forcing due to increases in tropospheric ozone are likely to be closer to our results. The global annual mean SEFDH forcing is 0.30 W m^{-2} , where 0.23 W m^{-2} is due to longwave forcing and 0.07 W m^{-2} is due to shortwave forcing. The global annual mean instantaneous forcing is 0.33 W m^{-2} . Note that up to 1970 the

change in column ozone is due to increased ozone in the troposphere. We have calculated the forcing from 1870 to 1970 and find a global annual mean forcing, assuming fixed dynamical heating, of 0.21 W m^{-2} . This value, in terms of the global mean, is in excellent agreement with the results of Stevenson *et al.* [1998].

4.2. Total Ozone

Of course, reduction in stratospheric ozone since 1970 has had a profound impact on the total column of ozone (compare Plates 1 and 3). This change in ozone distribution has a small negative effect on the radiative forcing. The geographic distribution of annual mean net SEFDH forcing for this time period is shown in Plate 5. The pattern of positive net forcing is similar to that for the tropospheric ozone forcing. The largest forcing in the Northern Hemisphere occurs at a latitude of 45° over the Atlantic and Pacific Oceans and central Europe and is due to shortwave forcing (see Plate 6a). The lower net forcing over the western United States and central Asia is due to a net negative longwave forcing in these regions (see Plate 6b). This negative forcing results from a colder stratosphere and decreased ozone. The decrease in ozone is largest in these regions, as pointed out in section 2. The large negative forcing over Antarctica is due to the ozone hole. The tropical forcing is positive and is dominated by the ozone increase associated with biomass burning. The global annual mean forcing is 0.29 W m^{-2} , with 0.18 W m^{-2} from the shortwave forcing and 0.11 W m^{-2} from longwave forcing. Thus stratospheric depletion has led to a -0.01 W m^{-2} change in the SEFDH forcing. The instantaneous net forcing is 0.41 W m^{-2} , which indicates the stratospheric adjustment has a significant effect on the forcing.

The change in the zonal annual mean stratospheric temperatures is shown in Figure 4. The largest decrease in temperature (-3.5 K) occurs over Antarctica associated with the ozone hole. Note that this value is less than that calculated assuming annual mean fixed dynamical heating, a result that agrees with the findings of Forster *et al.* [1997]. In the Northern Hemisphere lower stratosphere a cooling of less than -2 K occurs

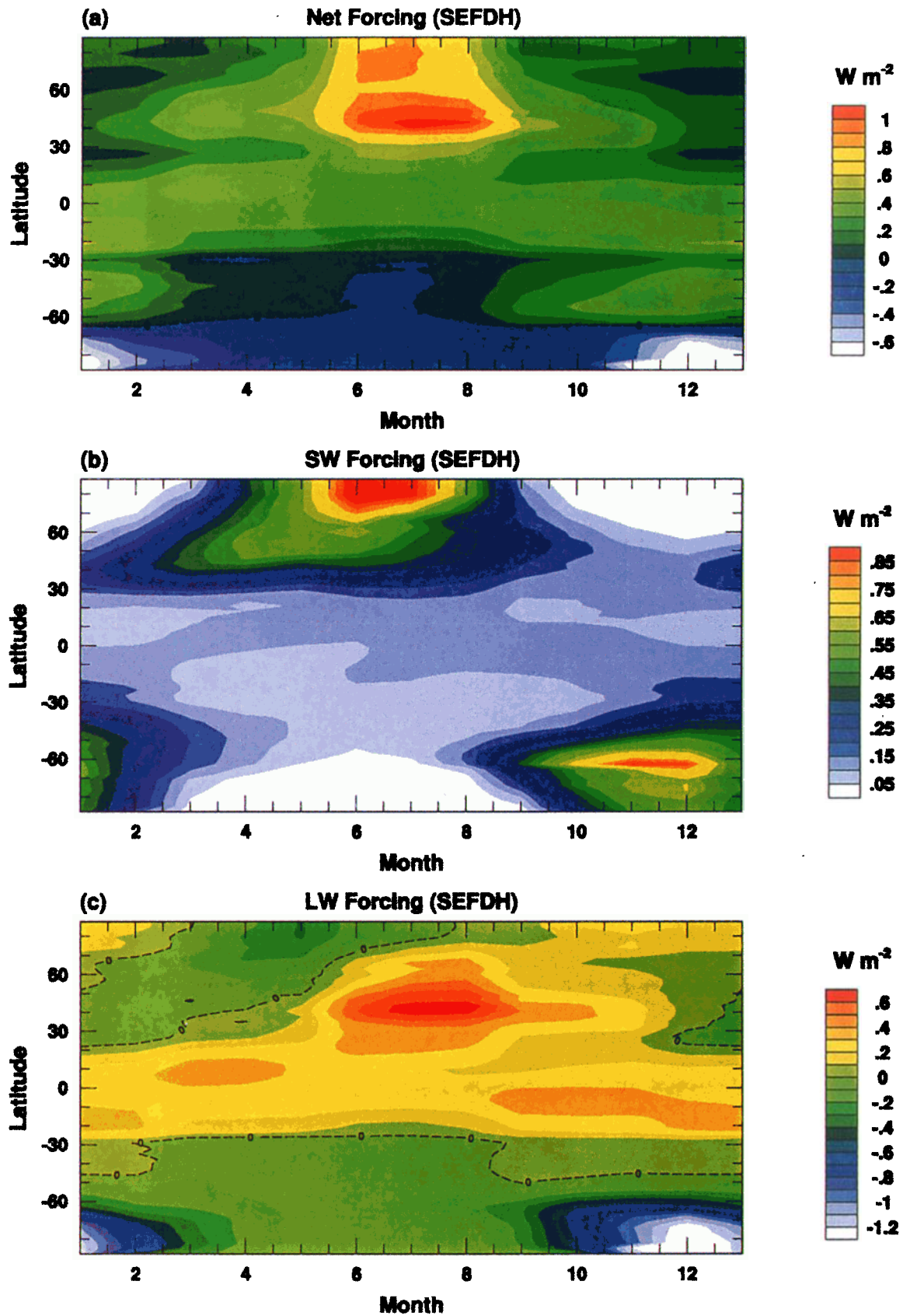


Plate 7. Seasonal cycle in the zonal mean SEFDH: (a) net, (b) shortwave, and (c) longwave forcing ($W m^{-2}$) due to total ozone changes from 1870 to 1990.

near 50°. Cooling in the tropical lower stratosphere is as low as -0.5 K, which is a result of increased tropospheric ozone leading to a decrease in longwave flux convergence into this region. A cooling in this region of -0.2 K occurs between 1870 and 1970, which indicates the important role that increased tropospheric ozone plays in determining the temperature change in this region. Cooling also occurs in both hemispheres just below 50 km. Kiehl and Boville [1988] have shown that the FDH approximation overestimates cooling at these altitudes (see, for example, their Figure 16). Thus dynamical response would tend to reduce the cooling in this region compared to what the SEFDH assumption predicts.

The seasonal cycle in the zonal mean SEFDH is shown in Plate 7a. Maximum positive forcing occurs between June and August at 45° north, with forcing as large as 1.0 W m^{-2} . There is also substantial forcing that occurs earlier in the year at the same latitude. Net negative forcing occurs throughout the year over Antarctica. The reason for these particular spatial-temporal patterns can be better understood by considering the shortwave (Plate 7b) and longwave (Plate 7c) seasonal forcing. The seasonal cycle in shortwave forcing indicates that the Northern Hemisphere springtime maximum in forcing is due to shortwave absorption in the troposphere. According to Plate 2, during this time, stratospheric ozone depletion reaches its highest level, while tropospheric ozone is increased. This implies that more shortwave radiation will be absorbed in the troposphere at this time of year. During Northern Hemisphere summer the maximum in forcing is a result of longwave processes (Plate 7c). Over Antarctica, longwave negative forcing due to the significant decrease in the downward flux from the stratosphere (equation (2)) leads to a net negative forcing throughout the year. Note that there is a narrow region of positive shortwave forcing in austral springtime, which is similar to the effect seen in the Northern Hemisphere.

Figure 5 shows the seasonal cycle in zonal mean SEFDH change in lower stratospheric temperature (i.e., one level above the tropopause). In the Northern Hemisphere maximum, cooling occurs from March to July between 30° and 90° north. This is due to the maximum level of ozone depletion at this time of year. The largest cooling, -2.5 K, occurs in May at 80° north. In the Southern Hemisphere the largest cooling occurs over Antarctica associated with the ozone hole [e.g., Kiehl et al., 1988]. Peak cooling (< -12 K) occurs in November at 75° south. Cooling in the tropics is uniform throughout the

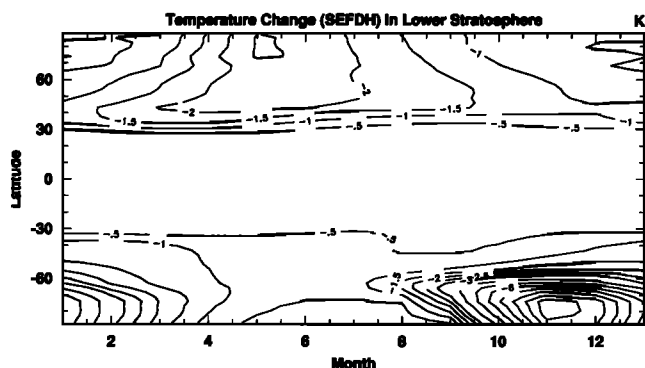


Figure 5. Seasonal cycle of the zonal mean change (1990–1870) in lower stratospheric temperature (K) from the SEFDH model. Contour interval is 1 K for temperature changes greater than 3 K and 0.5 K for changes less than 3 K.

Table 4. Forcing Due to Change in Tropospheric Ozone, Preindustrial to Present

Study	Forcing, (W m^{-2})
Forster et al. [1996]	0.30–0.51
Chalita et al. [1996]	0.28
Van Dorland et al. [1997]	0.38
Roelofs et al. [1997]	0.42
Haywood et al. [1998]	0.29–0.32
Stevenson et al. [1998]	0.29
Berntsen et al. [1998]	0.28–0.31
Brasseur et al. [1998]	0.37
Present work	0.30–0.33

year at a magnitude of -0.5 K or less. It must be noted that the large cooling in the Northern Hemisphere occurs in a region where we have the least confidence in the ozone data, i.e., poleward of 50°.

5. Summary and Conclusions

By combining a chemistry model, satellite data, and ozone-sonde data we have created a global hybrid data set for atmospheric ozone for 1870–1990. The unique aspect of this data, compared to results based solely on chemical transport models, is the use of the satellite data to constrain the column abundance and trends of ozone in the troposphere and total column. We have also used observed ozonesonde and Umkehr data to constrain the vertical profile of the ozone changes. These observational constraints result in ozone fields that are not totally model dependent. We have used the data to calculate climate forcing due to changes in tropospheric ozone and total column ozone between 1870 and 1990. A summary of our results compared to previous results for changes in tropospheric ozone is shown in Table 4. Our results are in good agreement with the latest studies of Bernsten et al. [1998] and Haywood et al. [1998]. Our results for 1990 tropospheric forcing are also in excellent agreement with Stevenson et al. [1998].

The magnitude of tropospheric forcing obviously depends on the assumed preindustrial ozone distribution. To better quantify the dependence on the preindustrial ozone amount, we have carried out two instantaneous forcing sensitivity studies. In the first study we assumed that the preindustrial ozone value was 5 ppbv at all tropospheric altitudes. The resulting forcing, based on this extreme assumption, is 1.55 W m^{-2} . This could be considered as an upper bound to tropospheric ozone forcing. In the second sensitivity calculation we subtracted a constant ozone amount (5.559 ppbv) from the 1870 profile. This preserves the vertical profile of ozone and insures that the global mean near-surface ozone mixing ratio is 10 ppbv. The forcing for this case is 0.49 W m^{-2} , which is 69% larger than our standard case. Thus the tropospheric forcing could potentially be larger if the preindustrial ozone was less than what we have calculated from the chemistry model. For both cases the maximum forcing occurred in the tropical and subtropical dry regions (e.g., Saharan region). This is due to two factors: the large difference between surface and tropopause temperature and minimal overlap between water vapor and ozone absorption.

The global annual mean SEFDH forcing due to total ozone change is 0.29 W m^{-2} , depending on the particular ozone data set used. This forcing is of the same magnitude but opposite in sign as that due to the direct effect of sulfate aerosols. There-

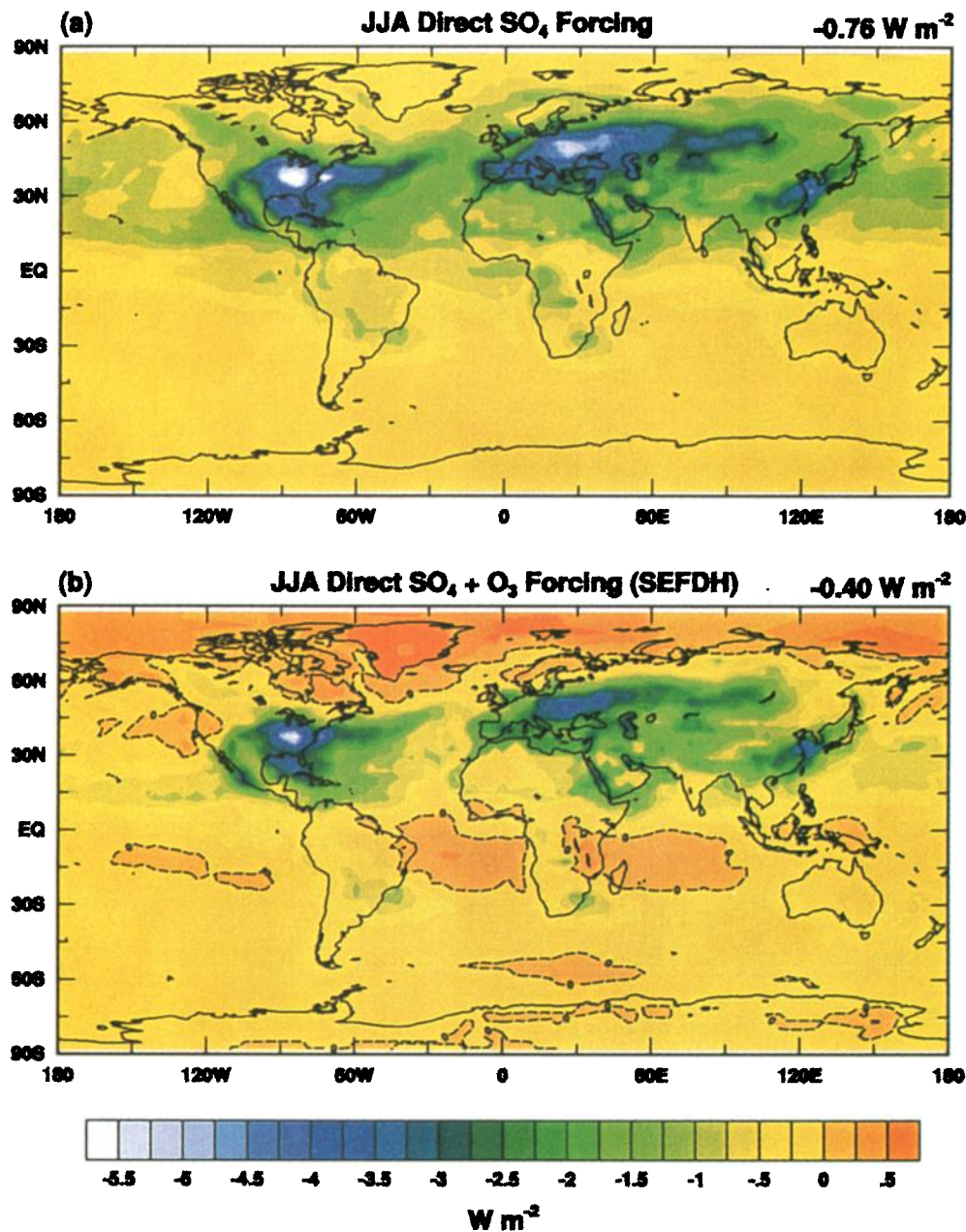


Plate 8. Geographic distribution in forcing (W m^{-2}) due to (a) direct effect of sulfate aerosols and (b) sulfate aerosols plus ozone (SEFDH) forcing for June-July-August for 1870 to 1990.

fore it is interesting to ask: to what extent the ozone forcing has offset the direct aerosol forcing due to sulfate aerosols? Kiehl *et al.* [1999] calculated the forcing due to sulfate aerosols from a version NCAR CCM3 that includes a sulfur chemistry model [Barth *et al.*, 1999; Rasch *et al.*, 1999]. The direct sulfate forcing for June-July-August from the CCM3 simulations is shown in Plate 8a. We choose these three months because they are when both sulfate direct forcing and tropospheric ozone forcing are largest. The largest sulfate forcings occur over regions of anthropogenic emission of sulfur dioxide, namely, the eastern United States, central Europe, and eastern China. The global mean for these three months is -0.76 W m^{-2} . The sum of the SEFDH ozone forcing plus the sulfate aerosol for June-July-August is shown in Plate 8b. At midlatitude regions of the Northern Hemisphere the negative forcing is reduced, espe-

cially over the Pacific and Atlantic Ocean regions. In the tropics the positive ozone forcing associated with biomass burning has more than offset the weak negative aerosol forcing. The global mean for the sum of aerosol forcing plus ozone forcing is -0.40 W m^{-2} . Thus the production of ozone by pollution reduces the direct sulfate aerosol forcing by 47%. Note that we have not accounted for aerosols associated with biomass burning, nor the indirect effect of sulfate aerosols. Both of these factors could act to counter the positive forcing due to increased ozone. It is important to note that the spatial pattern of forcing is quite sensitive to the role of ozone forcing.

Finally, the seasonal cycle of the zonal mean forcing due to the direct effect of sulfate aerosols plus greenhouse gases and including ozone forcing are shown in Plates 9a and 9b. The sulfate direct forcing plus greenhouse gases have a minimum

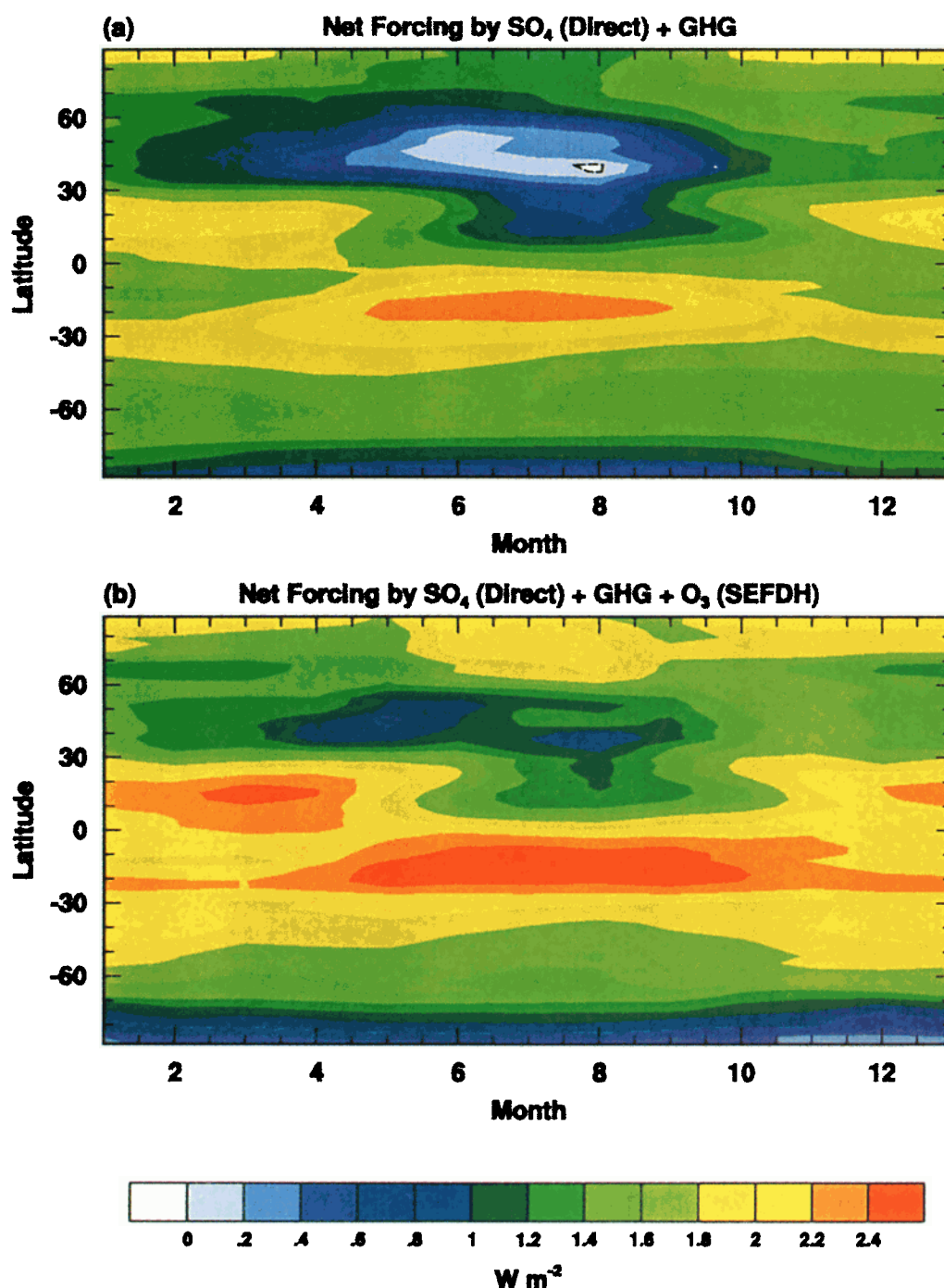


Plate 9. Seasonal cycle in zonal mean forcing (W m^{-2}) due to (a) direct effect of sulfate aerosols plus greenhouse gases and (b) direct effect of sulfate aerosols plus greenhouse gases plus ozone for SEFDH model. Changes are for 1870 to 1990.

between June and August at 30°–60° north. This minimum in total forcing is due to the strong negative forcing from sulfate aerosols. Maximum forcing of 2.3 W m^{-2} occurs in the tropics and is mainly due to the large surface-tropopause temperature difference in this region. This pattern indicates a strong meridional gradient in forcing between the tropics and the Northern Hemisphere. This gradient is largest between June and August. Since dynamical effects in the atmosphere are responsive to gradients in heating, the implication is that these effects may amplify in Northern Hemisphere summer due to anthropogenic activity. However, Plate 9b shows the seasonal cycle in

zonal forcing when ozone forcing is added to the effects of greenhouse gases and sulfate aerosols. The spatial and temporal pattern is significantly different due to ozone forcing. The minimum forcing between 30° and 60° north is $\sim 0.5 \text{ W m}^{-2}$, greater than 0.1 W m^{-2} without ozone forcing. Because of tropical ozone increases, associated with biomass burning, there is an increase in greenhouse forcing at 20° south. The change in meridional gradient of forcing is substantially different when ozone effects are included. These results indicate that ozone forcing plays an important role in the phase and meridional gradient of net climate forcing.

This study has focused on the radiative forcing due to changes in both tropospheric and stratospheric ozone. There are a number of areas where further work is needed to improve the ozone data sets. As stated in section 2, satellite data are not available poleward of $\sim 50^\circ$. We have extrapolated into these regions to create global data sets. Observations at high Northern Latitudes indicate significant ozone depletion in the region of the middle and upper troposphere [Tarasick *et al.*, 1995]. Thus one improvement would be to use in situ data at polar latitudes, instead of extrapolation. Another region that needs further investigation is the tropics. For our preindustrial ozone we have assumed that biomass burning is represented by 25% of the present-day biomass-burning-associated ozone in this region. Implicit in this assumption is that significant biomass burning occurred in 1870. If there was less burning [Wang and Jacob, 1998], then the forcing in this region would be larger than what we report. Finally, it would be of interest to use our data to carry out a more quantitative evaluation of ozone distributions produced by chemical transport models. The advantage of these models is that they allow us to calculate the chemical composition of the atmosphere for a wide range of conditions. They are therefore valuable tools in understanding the sensitivity of the chemistry of the Earth's atmosphere to various perturbations. They are also the only means we have to predict the future chemical composition of the atmosphere. Given the growing importance of these models for understanding chemistry-climate problems, it is imperative that they be evaluated for the present atmosphere.

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